

homework₁

June 12, 2026

1. If R is a ring, show that $M_n(R)$ is a ring under the usual operations of matrix addition and multiplication.
2. Let $(G, +)$ be an Abelian group. An **endomorphism** of G is a group homomorphism $\varphi : G \rightarrow G$. Let $\text{End}(G)$ denote the set of all endomorphisms on G . Given $\varphi, \psi \in \text{End}(G)$, define

$$(\varphi + \psi)(g) := \varphi(g) + \psi(g), \quad \text{and} \quad (\varphi \times \psi)(g) := \varphi(\psi(g)).$$

- (a) Show that $+$ and $\times$ are well-defined binary operations on $\text{End}(G)$.
 - (b) Show that $(\text{End}(G), +, \times)$ is a ring.
3. Let R be a ring and $a, b \in R$. Show that
 - (a) $0 \cdot a = a \cdot 0 = 0$.
 - (b) For any $n \in \mathbb{Z}$, $(na)b = n(ab) = a(nb)$. [Hint: Use Induction.]
 - (c) If R is unital, then the multiplicative identity is unique. Moreover, $(-1) \cdot a = (-a)$ for all $a \in R$.
 4. Let R be a commutative ring and let S and T be two subsets of R . Define

$$ST := \bigcup_{n=1}^{\infty} \left\{ \sum_{i=1}^n s_i t_i : s_i \in S, t_i \in T \right\}.$$

(In other words, elements in ST are finite sums of the form $\sum_{i=1}^n s_i t_i$, where $s_i \in S$ and $t_i \in T$, while the number n may vary.)

- (a) If S and T are both subrings of R , prove that ST is a subring.
 - (b) If S and T are both ideals, then prove that ST is an ideal.
5. Fix $m, n \in \mathbb{Z}$ and let $d := \gcd(m, n)$. If I denoted the ideal generated by $\{m, n\}$, then show that $I = (d)$, the principal ideal generated by d (See Definition 5.3).

Extra Problems

You need not submit these problems; they are here only for practice.

****Definition:**** A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to have ****compact support**** if there exists $a, b \in \mathbb{R}$ such that $f(x) = 0$ for all $x \notin [a, b]$.

Define

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \rightarrow \mathbb{C} : f \text{ is continuous and has compact support}\}.$$

Equip $C_c(\mathbb{R})$ with the pointwise operations as in Part (v) of Example 1.2 in the notes. You need not check that $C_c(\mathbb{R})$ is a ring (this is easy to do).

1. Prove that $C_c(\mathbb{R})$ is not unital.

2. Let R be a ring, S be a subring of R and let I be an ideal of R .
- (a) Prove that $S + I := \{s + a : s \in S, a \in I\}$ is a subring of R .
 - (b) Prove that $I \triangleleft S + I$.
 - (c) Prove that $S \cap I \triangleleft S$.
 - (d) Prove that

$$\frac{S + I}{I} \cong \frac{S}{S \cap I}$$

[Hint: Define $\varphi : S \rightarrow (S + I)/I$ by $s \mapsto s + I$.]

3. Let $D \in \mathbb{Q}$ be a rational number that is not a perfect square. Define

$$R := \{a + b\sqrt{D} : a, b \in \mathbb{Q}\}.$$

- (a) Show that R is a subring of $\mathbb{C}$.
 - (b) If $(a + b\sqrt{D}) \neq 0$, then show that $a^2 - b^2D \neq 0$.
 - (c) Use Part (b) to prove that R is a field. This field is denoted $\mathbb{Q}(\sqrt{D})$, and is called a **quadratic field**.
4. For a unital ring R , define the **center** of R to be

$$\mathcal{Z}(R) := \{a \in R : ab = ba \text{ for all } b \in R\}.$$

Show that $\mathcal{Z}(R)$ is a subring of R .

Given the content in the image, —

1. Given: R is a ring. Show that $M_n(\mathbb{C}R)$ is a ring under $C+$, matrix multiplication.

Definition 0.1. Fix $n \in \mathbb{N}$. Show $M_n(\mathbb{C}R)$ is a ring under addition and multiplication of real numbers is $C = AB$ such that

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Proof. **Associativity of multiplication:**

$$A \cdot (B \cdot C) = A \cdot D \quad \text{where} \quad d_{ij} = \sum_{k=1}^n b_{ik}c_{kj}.$$

$$\begin{aligned} \Rightarrow A \cdot (B \cdot C) &= \left(\sum_{k=1}^n a_{ik}d_{kj} \right)_{n \times n} \\ &= \left(\sum_{k=1}^n a_{ik} \left(\sum_{l=1}^n b_{kl}c_{lj} \right) \right)_{n \times n} \\ &= \left(\sum_{k=1}^n \sum_{l=1}^n a_{ik}b_{kl}c_{lj} \right)_{n \times n} \end{aligned}$$

$$= (A \cdot B) \cdot C.$$

Distributive Property:

$$\begin{aligned} AB + AC &= \left(\sum_{k=1}^n a_{ik}b_{kj} + \sum_{k=1}^n a_{ik}c_{kj} \right)_{n \times n} \\ &= \left(\sum_{k=1}^n a_{ik}(b_{kj} + c_{kj}) \right)_{n \times n} \\ &= A(B + C). \end{aligned}$$

Similarly, we can show $(A + B)C = AC + BC$. □

2.

Theorem 0.1. Conclude that if R is a ring, then $M_n(R)$ is a ring.

—

Given $(G, +)$ is an Abelian group,

$$\text{End}(G) = \{\varphi : G \rightarrow G \mid \varphi \text{ is a group homomorphism}\}$$

For $\varphi, \psi \in \text{End}(G)$,

$$\begin{aligned} (\varphi + \psi)(g) &= \varphi(g) + \psi(g) \\ (\varphi \times \psi)(g) &= \varphi(\psi(g)) \end{aligned}$$

(a) Show the operations given above are well defined.

Proof. We need to show that $\varphi + \psi \in \text{End}(G)$. Consider $g, h \in G$:

$$(\varphi + \psi)(g + h) = \varphi(g + h) + \psi(g + h) = \varphi(g) + \varphi(h) + \psi(g) + \psi(h)$$

Using the abelian property here,

$$= \varphi(g) + \psi(g) + \varphi(h) + \psi(h) = (\varphi + \psi)(g) + (\varphi + \psi)(h)$$

Thus, $\varphi + \psi \in \text{End}(G)$.

We also need to show that $\varphi \times \psi \in \text{End}(G)$. Consider $g, h \in G$:

$$\begin{aligned} (\varphi \times \psi)(g + h) &= \varphi(\psi(g + h)) = \varphi(\psi(g) + \psi(h)) = \varphi(\psi(g)) + \varphi(\psi(h)) \\ &= (\varphi \times \psi)(g) + (\varphi \times \psi)(h) \end{aligned}$$

Thus, $\varphi \times \psi \in \text{End}(G)$. □

(b) Prove $(\text{End}(G), +, \times)$ is a ring.

Proof. As G is an abelian group, $(\text{End}(G), +)$ is an abelian group.

Associativity holds as composition of functions is associative.

Distributive Property:

$$\varphi \times (\psi_1 + \psi_2)(g) = \varphi \times (\psi_1 + \psi_2)(g)$$

$$= \varphi(\psi_1(g) + \psi_2(g)) = \varphi(\psi_1(g)) + \varphi(\psi_2(g))$$

$$= (\varphi \times \psi_1)(g) + (\varphi \times \psi_2)(g) = (\varphi \times \psi_1 + \varphi \times \psi_2)(g)$$

Thus, $(\text{End}(G), +, \times)$ is a ring. □

$$(\varphi_1 + \varphi_2) \times \varphi(g) = (\varphi_1 + \varphi_2)(\varphi(g)) = \varphi_1(\varphi(g)) + \varphi_2(\varphi(g)) = \varphi_1 \times \varphi(g) + \varphi_2 \times \varphi(g).$$

1. Given: $(R, +, \cdot)$ is a ring; $a, b \in R$

a) Show: $0 \cdot a = a \cdot 0 = 0$

$$\textit{Proof. } 0 \cdot a = (a - a) \cdot a = a^2 - a^2 = 0$$

$$a \cdot 0 = a \cdot (a - a) = a^2 - a^2 = 0$$

□

b) Show: $(na)b = n(ab) = a(nb)$

Proof. Base case ($n = 1$): Trivial.

Induction Hypothesis: $(na)b = n(ab) = a(nb)$

For $n + 1$ times:

$$(a + \cdots + a + n + 1 \text{ times})b = (na + a)b = nab + ab = (n + 1)ab$$

$$a(n + 1 \text{ times } b) = a(n \text{ times } b + b) = nab + ab$$

□

c) If R is unital, prove: 1 is multiplicative identity is unique.

Proof. Assume $1_A = 1_B$ such that $1_A \cdot 1_B = 1_B \Rightarrow$ Multiplicative identity is unique.

Show: $(-1) \cdot a = -a$ for any $a \in R$

$$\textit{Proof. } a + (-1)a = 1 \cdot a + (-1)a = (1 + (-1))a = 0$$

$$\Rightarrow (-1)a \text{ is the additive inverse} \Rightarrow (-1) \cdot a = -a$$

□

□

2. Given: R is commutative ring, $S, T \subseteq R$

$$ST := \bigcup_{n=1}^{\infty} \left\{ \sum_{i=1}^n s_i t_i \mid s_i \in S, t_i \in T \right\}$$

a) If S and T are both subrings, show: ST is a subring.

Proof. Assume S and T to be non-empty. □

Let $a, b \in ST$ such that $a = \sum_{i=1}^{n_1} s_i t_i$ and $b = \sum_{i=1}^{n_2} s'_i t'_i$.

Without loss of generality, assume $n_2 > n_1$. Then,

$$b - a = \sum_{i=1}^{n_2} s'_i t'_i - \sum_{i=1}^{n_1} s_i t_i = \sum_{i=n_1+1}^{n_2} s'_i t'_i + \sum_{i=1}^{n_1} (s'_i t'_i - s_i t_i).$$

By the Abelian property, we can rearrange the terms to get

$$\sum_{i=n_1+1}^{n_2} s'_i t'_i + \sum_{i=1}^{n_1} (s'_i t'_i - s_i t_i) = \sum_{i=1}^{n_2} s''_i t''_i.$$

Since $i = 1$ to $i = n_1$ is set to 0 for the first entry, we have

$$(ST, +) \prec (R, +).$$

Proof. Check the ring property:

$$\left(\sum_{i=1}^{n_1} s_i t_i \right) \left(\sum_{j=1}^{n_2} s'_j t'_j \right).$$

Without loss of generality, assume $n_1 < n_2$. Now set all terms to zero in the first entry after $i > n_1$,

$$\left(\sum_{i=1}^{n_2} s_i t_i \right) \left(\sum_{j=1}^{n_2} s'_j t'_j \right).$$

Every term will be of the form $s_i s'_j t_i t'_j$ for some $1 \leq i, j \leq n_2$ and as R is a commutative ring, we get $s_i s'_j t_i t'_j$ and as S, T are subrings, $s_i s'_j \in S$ and $t_i t'_j \in T$ respectively. □

Definition 0.2. If S, T are both ideals, then ST is an ideal.

Proof. Consider $\sum_{i=1}^n s_i t_i \in ST$ and $r \in R$. Then,

$$r \left(\sum_{i=1}^n s_i t_i \right) = \sum_{i=1}^n (r s_i) t_i.$$

Since S is an ideal, we get $rs_i \in S$ for all $i \leq n$. Thus,

$$r \left(\sum_{i=1}^n s_i t_i \right) = \sum_{i=1}^n s'_i t_i \in ST.$$

Similarly,

$$\left(\sum_{i=1}^n s_i t_i \right) r = \sum_{i=1}^n s_i (t_i r).$$

Since T is an ideal, we get $t_i r \in T$ for all $i \leq n$. Thus,

$$\left(\sum_{i=1}^n s_i t_i \right) r = \sum_{i=1}^n s_i t'_i \in ST.$$

□

—

Given $m, n \in \mathbb{Z}$ and $d = \gcd(m, n)$,

Definition 0.3. Let I be the ideal generated by $\{m, n\}$.

Theorem 0.2. Show that $I = (d)$.

Proof. Let $\mathcal{F} = \{J \subseteq \mathbb{Z} \mid J \text{ is an ideal and } X \subseteq J\}$, where $X = \{m, n\}$.

Define $(X) = \bigcap_{J \in \mathcal{F}} J$.

If $a \in (X)$, then $a \in J$ for every $J \in \mathcal{F}$.

Since J is an ideal of $\mathbb{Z}$, it must also be a subgroup of $\mathbb{Z}$, so there exists $n_J \in \mathbb{Z}$ such that $J = n_J \mathbb{Z}$.

If $x \in J$, then $n_J \mid x$ and $n_J \mid y$ for some $x, y \in X$.

Since $d = \gcd(x, y)$, we have $d \mid n_J$.

Thus, $n_J \mathbb{Z} \subseteq d\mathbb{Z}$, where n_J is the integer corresponding to J .

Therefore, $(X) \subseteq d\mathbb{Z}$.

$d\mathbb{Z}$ is the smallest ideal containing X .

Consider $n\mathbb{Z} \subseteq \mathbb{Z}$ such that $X \subseteq n\mathbb{Z}$ and $n\mathbb{Z} \subseteq d\mathbb{Z}$.

Let $nk_1 = x$ and $nk_2 = y$.

By Bézout's identity, $nk_1 r_1 + nk_2 r_2 = xr_1 + yr_2 = d$ as $d = \gcd(x, y)$ and $k_1, k_2, r_1, r_2 \in \mathbb{Z}$.

Thus, $n(k_1 r_1 + k_2 r_2) = d \implies n \mid d$.

Hence, $d\mathbb{Z} \subseteq n\mathbb{Z} \implies n\mathbb{Z} = d\mathbb{Z}$.

□

—

Given the content in the image, —

Definition 0.4. Let $C_c(\mathbb{R})$ be the set of continuous functions $f : \mathbb{R} \rightarrow \mathbb{C}$ with compact support, i.e.,

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is continuous and } \exists a, b \in \mathbb{R} \text{ such that } f(x) = 0 \text{ whenever } x \notin [a, b]\}.$$

The operations $+$ and $\times$ are defined as usual addition and multiplication of functions.

Remark 0.1. Observe that $C_c(\mathbb{R})$ is a ring and is non-empty.

Theorem 0.3. Show that $C_c(\mathbb{R})$ is not unital.

Proof. Assume there is a $g \in C_c(\mathbb{R})$ such that $f \times g = f$ for any $f \in C_c(\mathbb{R})$.

If $g(x_0) \neq 0$ for some $x_0 \in [a, b]$, choose $f \in C_c(\mathbb{R})$ such that $f(x) \neq 0$ for $x \in [a', b']$ where $[a, b] \subset [a', b']$.

Consider $x_0 = \frac{a'+a}{2}$. Then,

$$g(x_0) = 0 \quad \text{and} \quad f(x_0) \neq 0 \implies f \times g(x_0) \neq 0.$$

This is a contradiction. Hence, no unit exists in $C_c(\mathbb{R})$. □

Given the content in the image, —

1. Given: R is a ring, S be a subring of R and $I \trianglelefteq R$.

(a)

Theorem 0.4. Prove: $S + I := \{s + a \mid s \in S, a \in I\}$ is a subring.

Proof. Observe, $S + I$ is non-empty. Then consider $a', b' \in S + I$ where $a' = s_1 + a$ and $b' = s_2 + b$. Now,

$$a' - b' = (s_1 + a) - (s_2 + b) = (s_1 - s_2) + (a - b) \in S + I,$$

where $s_1 - s_2 \in S$ and $a - b \in I$. Thus, $(S + I, +) \leq (R, +)$.

Now, let $a', b' \in S + I$ where $a' = s_1 + a$, $b' = s_2 + b$. Then,

$$a' \cdot b' = (s_1 + a)(s_2 + b) = s_1 s_2 + a s_2 + s_1 b + ab,$$

where $s_1 s_2 \in S$ and $a s_2 + s_1 b + ab \in I$. Therefore, $(S + I, +, \cdot)$ is a subring of $(R, +, \cdot)$. □

(b)

Theorem 0.5. Prove: $I \trianglelefteq S + I$.

Proof. Consider $a \in I$ and $b' \in S + I$ such that $b' = s + b$ where $s \in S$, $b \in I$. Observe

$$ab' = a(s + b) = as + ab,$$

as $I \trianglelefteq R$, $as \in I$ and $ab \in I \implies ab' \in I$. Similarly, $b'a \in I$. □

(c)

Theorem 0.6. Show: $S \cap I \trianglelefteq S$.

Proof. As $(I, +) \trianglelefteq R$,

$$(S \cap I, +) \trianglelefteq S \quad \text{and} \quad 0 \in S \cap I \implies S \cap I \neq \emptyset.$$

If $a \in S \cap I$ and $s \in S$, then $a \in I$ and $s \in S \implies as \in I$. Now, $a \in S$ and $s \in S \implies as \in S$. Thus, $S \cap I \trianglelefteq S$. $\square$

Proof. Hence, as $s \in S \cap I$ ($\iff \forall s \in S \cap I$), we have $S \cap I \triangleleft S$.

(cd) Prove: $\frac{S+I}{I} \cong \frac{S}{S \cap I}$

Proof:

(1) Define $\varphi : S \rightarrow \frac{S+I}{I}$ by $\varphi(s) = s + I$ (i.e., $s \mapsto s + I$).

Well-definedness:

$$s_1 = s_2 \implies s_1 + I = s_2 + I.$$

(2) φ is a ring homomorphism:

$$\varphi(s + s') = s + s' + I = (s + I) + (s' + I) = \varphi(s) + \varphi(s'),$$

$$\varphi(ss') = ss' + I = (s + I)(s' + I) = \varphi(s)\varphi(s').$$

(3) φ is surjective (by construction).

(4) $\ker \varphi = S \cap I$. If $s \in S \cap I$, then $\varphi(s) = 0 + I$ as $s \in S \cap I$. Hence, $S \cap I \subseteq \ker \varphi$. If $s \in \ker \varphi$, $\varphi(s) = I$, then $s \in S$ and $s \in I$,
 $\implies s \in S \cap I$.

Thus, $S \cap I = \ker \varphi$.

By the First Isomorphism Theorem,

$$\frac{S+I}{I} \cong \frac{S}{S \cap I}.$$

$\square$

Given $D \in \mathbb{Q}$ be a rational number and is NOT a perfect square.

$$R := \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

(a)

Proof. Observe, $R \neq \emptyset$ as $0 \in R$.

Consider $x, y \in R$, where $x = a_1 + b_1\sqrt{D}$ and $y = a_2 + b_2\sqrt{D}$; $a_1, b_1, a_2, b_2 \in \mathbb{Q}$.

$$x - y = (a_1 - a_2) + (b_1 - b_2)\sqrt{D} \in R \text{ as } a_1 - a_2 \in \mathbb{Q} \text{ and } b_1 - b_2 \in \mathbb{Q}.$$

Thus, $(R, +)$ is a subgroup of $(\mathbb{C}, +)$.

Now, look at $xy = (a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D})$

$$= a_1a_2 + b_1b_2D + (a_2b_1 + a_1b_2)\sqrt{D}, \text{ and} \\ a_1a_2 + b_1b_2D, \quad a_2b_1 + a_1b_2 \in \mathbb{Q}.$$

Hence, $xy \in R$ and $(R, +, \cdot)$ is a subring of $(\mathbb{C}, +, \cdot)$. □

(b)

Theorem 0.7. If $(a + b\sqrt{D}) \neq 0$, then show that $a^2 - b^2D \neq 0$.

Proof. If $(a + b\sqrt{D}) \neq 0$ then $a \neq 0$ and $b \neq 0$.

Hence, $a - b\sqrt{D} \neq 0$. Now, $(a + b\sqrt{D})(a - b\sqrt{D}) \neq 0$

$$\Rightarrow a^2 - b^2D + 0(\sqrt{D}) \neq 0.$$

□

(c)

Theorem 0.8. Prove: R is a field.

Proof. Observe R is a commutative, unital ring.

Moreover, R is a division ring as for $a + b\sqrt{D} \neq 0$ then $\frac{a-b\sqrt{D}}{a^2-b^2D}$ is the inverse with respect to multiplication.

Now, R is a commutative, division ring, which is a field. □

Given the content in the image, —

Definition 0.5. Let R be a unital ring. The *center* of R , denoted $Z(R)$, is defined as

$$Z(R) = \{a \in R \mid ab = ba \text{ for all } b \in R\}.$$

Proof. To show that $Z(R)$ is a subring of R , we proceed as follows:

First, observe that $Z(R) \neq \emptyset$ since $1 \in Z(R)$.

Now, consider $a, a' \in Z(R)$. For any $b \in R$, we have

$$(a - a')b = ab - a'b = ba - ba' = b(a - a'),$$

which implies that $(Z(R), +)$ is a subgroup of R .

For any $b \in R$, observe that

$$(aa')b = a(a'b) = a(ba') = (ab)a' = (ba)a' = b(aa'),$$

which implies that $aa' \in Z(R)$.

Hence, $Z(R)$ is a subring of R . □

